DO NOW — RECALL

# Tracing unfamiliar algorithms

Name: _____ Class: _____ Date: _____

**1. A half adder receives 1 and 1. Give Sum and Carry.**

**2. What stays in a simplified expression after grouping four 1s?**

**3. How many items are in int[] a={2,3,5,8}?**

**4. Starting x=0, execute x+=3 three times. List x after each.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Sum=0, Carry=1.

### 2. Model answer

Only variables that are constant across the group.

### 3. Model answer

4.

### 4. Model answer

3, 6, 9.

### Vocabulary — know these terms

trace table — Records variable values at successive execution steps.

accumulator — Variable storing a running total.

counter — Variable tracking how often an event occurs.

conditional — Instruction executed only when a Boolean test is satisfied.

iteration — Repeated execution of a statement block.

NEW LEARNING

# Read, annotate and explain

### Choose trace columns

Track i, a[i] and total. Start with total=0.

### Check the condition

Odd values 3 and 1 do not satisfy a[i] MOD 2 = 0; total stays 0.

### Update then finish

4 and 2 are even, so total becomes 4 then 6; the printed output is 6.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Trace the sum of numbers greater than 3 in [2,5,4].

### Predict and explain

Create columns index, value and total; begin at 0.

### Check and extend

2 is skipped, then 5 is added.

### Check and extend

4 is also added. Final total is 9.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Trace total=0; for values [2,3,4], add each value. Give total after each.

### Question 2

For [4,7,2,6], count values greater than 5. What is returned?

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

An algorithm intends to count even numbers but uses if n%2==1. Identify the error and correct it.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**When tracing a loop, you can ignore the condition and update the accumulator every iteration.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.